

WAINWRIGHT CFB, AB, Canada

WMO: 711180

Lat: 52.8305N

Lon: 111.0957W

Elev: 2251

StdP: 13.54

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-20.7	-33.3	0.9	-26.0	-27.6	1.3	-20.6	25.3	27.8	22.4	30.0	6.2	230	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	85.4	62.9	81.8	61.9	78.4	60.4	66.2	79.2	64.3	77.1	62.6	74.6	10.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	90.5	70.5	60.0	83.9	68.7	57.9	77.7	67.3	32.1	78.4	30.7	77.2	29.3	74.7	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	19.1	16.7	DB	-33.5	92.9	6.0	4.2	-37.8	95.9	-41.3	98.4	-44.7	100.7	-49.0	103.8	(4)
(5)			WB	-33.7	69.3	5.9	2.3	-37.9	70.9	-41.4	72.3	-44.8	73.6	-49.1	75.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	9.2	12.3	23.1	39.6	51.4	59.0	64.3	62.2	53.3	39.9	24.3	11.3		
	DBStd	22.75	16.46	13.95	14.40	10.05	8.24	5.99	5.59	6.17	8.07	9.37	12.15	14.73		
	HDD50	5934	1266	1057	834	329	82	6	0	2	56	330	772	1200		
	HDD65	10118	1731	1477	1299	761	427	196	80	126	356	778	1222	1665		
	CDD50	1415	0	0	0	19	124	276	444	379	156	17	0	0		
	CDD65	122	0	0	0	0	4	15	59	38	6	0	0	0		
	CDH74	1909	0	0	0	6	118	238	739	607	197	4	0	0		
	CDH80	509	0	0	0	0	19	48	218	170	54	0	0	0		
	WSAvg	8.0	7.8	7.6	8.3	9.2	8.8	7.8	7.5	7.2	7.8	8.4	7.9	7.6		
	PrecAvg	16.1	0.8	0.5	0.7	1.2	1.7	2.9	2.8	2.2	1.3	0.7	0.7	0.7		
	PrecMax	21.5	2.4	0.9	1.7	2.8	4.1	4.3	5.9	4.9	3.3	1.9	2.4	1.2		
(15) Precipitation	PrecMin	10.3	0.2	0.1	0.0	0.2	0.4	0.7	0.7	0.2	0.0	0.2	0.1	0.1		
	PrecStd	3.3	0.5	0.3	0.4	0.6	0.9	1.0	1.2	0.9	0.9	0.4	0.5	0.3		
	DB	42.4	42.4	56.6	73.1	82.7	84.8	90.8	91.0	86.7	72.9	54.5	45.6			
	MCWB	36.0	36.2	44.9	52.4	57.3	61.8	65.6	64.4	60.5	54.4	42.9	37.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	39.0	38.4	49.3	67.7	77.2	80.1	85.7	84.4	79.3	64.5	48.0	39.9			
	2% MCWB	34.0	33.3	39.4	49.1	55.0	60.0	64.5	63.0	58.0	50.5	39.7	34.5			
	5% DB	35.5	34.8	44.0	61.6	72.7	76.3	81.7	80.6	74.6	59.1	43.3	35.0			
	5% MCWB	31.5	31.0	36.9	45.8	53.3	58.4	63.6	62.2	56.6	47.0	36.8	30.9			
	10% DB	31.2	31.4	40.1	56.6	67.9	72.3	77.8	76.8	69.1	54.2	39.1	31.0			
	10% MCWB	28.0	28.3	34.5	43.4	50.8	56.9	62.1	60.8	54.2	44.1	34.3	27.9			
	0.4% WB	36.5	36.8	45.0	53.1	59.8	65.3	69.6	68.0	62.1	55.0	43.9	38.3			
	0.4% MCDB	41.6	42.1	55.9	70.9	75.7	78.4	81.0	80.4	78.1	70.6	52.6	44.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.2	33.8	40.2	49.8	56.7	62.5	67.1	65.6	59.6	51.3	40.5	34.5			
	2% MCDB	38.6	37.8	47.5	65.9	73.2	74.5	80.1	78.5	76.4	61.8	47.4	39.5			
	5% WB	31.7	31.2	37.2	46.7	54.4	60.5	65.2	63.7	57.4	48.1	37.4	31.2			
	5% MCDB	35.5	34.7	43.4	60.0	69.4	71.7	77.9	76.6	72.1	58.2	42.4	34.9			
	10% WB	28.1	28.2	34.8	44.1	52.3	58.5	63.5	61.9	55.1	45.1	34.8	27.9			
	10% MCDB	31.0	31.2	39.7	54.9	65.8	69.5	75.3	74.3	67.3	53.1	38.8	30.8			
	MDBR	16.5	17.6	17.4	20.7	22.5	20.4	21.7	22.9	23.0	19.2	15.5	15.6			
	5% DB	18.1	20.0	21.6	29.5	29.5	27.5	28.7	30.9	33.3	28.2	20.4	18.6			
(36) Mean Daily Temperature Range	5% DB	14.6	16.2	14.0	15.4	13.2	12.2	11.9	12.8	14.9	15.4	13.8	14.6			
	5% WB	18.4	19.5	20.3	27.4	26.6	23.1	25.1	26.3	30.2	26.4	19.4	18.6			
	MCWBR	15.1	16.1	13.6	15.0	12.6	11.4	11.7	12.0	14.2	15.0	13.6	15.0			
	MCWBR	15.1	16.1	13.6	15.0	12.6	11.4	11.7	12.0	14.2	15.0	13.6	15.0			
(40) Clear-Sky Solar Irradiance	taub	0.254	0.295	0.290	0.333	0.354	0.371	0.364	0.378	0.317	0.280	0.256	0.234			
	taud	2.319	2.227	2.404	2.353	2.346	2.329	2.383	2.321	2.461	2.540	2.450	2.357			
	Ebn at Noon	239	257	288	286	284	279	279	268	275	267	241	229			
	Edh at Noon	18	28	29	35	37	38	35	36	27	20	16	15			
(44) All-Sky Solar Radiation	RadAvg	337	671	1139	1500	1839	1891	1986	1616	1139	674	337	252			
	RadStd	31	50	87	139	126	133	101	128	121	92	35	23			

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page